

huestegge-and-philipp-2011-effects-of-spatial-compatibili

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Paper Synthesis

Paper:

huestegge-and-philipp-2011-effects-of-spatial-compatibility-on-integration-processes-in-graph-comprehension

Executive Summary

The paper investigates how spatial compatibility between a graph's data region and its legend affects integration processes during graph comprehension. Across four experiments, compatible legend layouts consistently reduced response times, improved accuracy (for bar graphs), and lowered gaze transitions, especially for interaction judgments and under higher visual complexity, indicating that spatial alignment facilitates integration. The authors conclude that spatial compatibility is a key determinant of integration efficiency in graph reading.

Reading Recommendation

Read Selectively

The paper offers novel eye-tracking evidence that spatial compatibility between data regions and legends can speed graph judgments and reduce integration difficulty, especially for interaction judgments and under higher visual complexity. However, methodological concerns (confounded layout manipulations, small under-powered samples, incomplete reporting of Experiment 4) limit the robustness of the claims. Readers should focus on the experimental methods, key RT and eye-movement results, and the discussion of theoretical implications, while skimming the broader introductory and background sections.

Background and Motivation

The paper addresses the largely overlooked narrow-sense spatial (S-R and S-S) compatibility between a graph's data region and its legend, a factor whose contribution to the integration phase of graph comprehension has not been empirically isolated. Prior work on spatial compatibility (e.g., Proctor & Vu 2006; Hommel & Prinz 1997; Kornblum 1992) established performance benefits of compatible stimulus-response mappings, while Carpenter and Shah's (1998) three-phase model highlighted integration as a distinct stage but left its spatial determinants under-examined. Feeny et al. (2000) and Renshaw et al. (2004) provided initial eye-movement evidence for legend-related spatial effects, yet did not link these to the integration process per se. Thus, the study fills the gap by testing whether data-legend spatial compatibility directly impedes integration, especially under varying graph complexity.

Theoretical Stance

Overall stance: Consistent Theoretical

The introduction foregrounds spatial compatibility as the key determinant of integration, and the discussion repeatedly reaffirms this claim, extends it to relative proximity, and interprets the empirical findings (RTs, accuracy, gaze transitions) as evidence for the compatibility-based account, showing alignment across sections.

Theories Discussed

- **Spatial compatibility (S-R and S-S compatibility) (*focal*)**: The paper's central theoretical claim that spatial alignment between the data region and legend facilitates integration processes in graph comprehension.
- **Carpenter & Shah three-phase model of graph comprehension (*background*)**: A model distinguishing encoding, interpretation, and integration phases; used to locate the hypothesized effect of spatial compatibility on the integration phase.

- **General spatial compatibility literature (Proctor & Vu 2006; Hommel & Prinz 1997; Kornblum 1992)** (*background*): Broad findings that compatible stimulus-response mappings reduce errors and response times, providing a theoretical foundation for the focal claim.
- **Surface-information view** (*alternative*): A competing account suggesting that legend information is stored as surface codes and that spatial position of legend entries should not affect integration.
- **Proximity compatibility principle (Wickens & Carswell 1995; Wickens & Hollands 2000)** (*background*): The idea that absolute spatial proximity of display elements aids performance; the authors extend it to relative spatial proximity between legend and data.
- **Gillan et al. legend-indicator ordering prediction** (*alternative*): The claim that the order of legend symbols should match the order of indicators in the data region; the paper tests this prediction.
- **Optimal scan-path assumption** (*alternative*): The notion that an optimal eye-movement path includes only necessary transitions; the authors argue against it by invoking working-memory limits.
- **Expectation of spatial layout** (*background*): The hypothesis that participants form expectations about spatial layout, influencing integration difficulty when expectations are violated.

Study Design

- **Number of experiments:** 4
- **Predictors examined:** spatial compatibility (compatible vs. incompatible), effect type (main effect vs. interaction), match (matched vs. mismatched) - used in Experiment 3
- **Outcomes examined:** manual reaction time (RT), accuracy (percentage correct), gaze transition frequency between data region and legend
- **Materials:** Bar graphs (four bars, gray/black) with legends placed above the data region; grayscale line graphs (two uncrossed lines) with legends on the right side; more complex line graphs (six lines with varied markers) and legends of matching size; all stimuli were generated in-house (Excel/graphics software) and varied in spatial compatibility by aligning or misaligning legend entries with the corresponding data elements.
- **Design types:**
 - exp_1: within-subjects 2 (compatibility) $\times$ 2 (effect type) factorial; 18 participants, 64 trials per participant.

- exp_2: within-subjects 2 (compatibility) $\times$ 2 (effect type) factorial; 16 participants, 64 trials per participant; same structure as exp_1 but with line graphs.
- exp_3: within-subjects 2 (compatibility) $\times$ 2 (statement type) $\times$ 2 (match) factorial; 18 participants, 64 trials per participant; increased visual complexity with six-line graphs and larger legends.
- exp_4: memory-based task using line graphs; compatibility manipulation present but design details (predictor levels, counterbalancing) not fully specified; 12 participants, fixed 5 s graph exposure followed by statement verification.

Findings Summary

Across the four experiments, spatial incompatibility between the data region and legend consistently increased response times, reduced accuracy (significant only for bar graphs), and elevated the number of gaze transitions, indicating more demanding integration processes. The compatibility effect was strongest for interaction judgments in the simpler bar and line graphs (Experiments 1 and 2) and only emerged for main-effect judgments when visual complexity was heightened in the more elaborate line graphs (Experiment 3). Experiment 4 showed that the incompatibility disadvantage persisted even when participants had to retrieve graph information from memory rather than view the graph directly. However, the effect was absent for main-effect RTs in Experiments 1-2 and for accuracy in Experiment 2, suggesting that the impact of incompatibility depends on task and stimulus complexity.

Evidence-Claim Alignment

Supported Claims

- Spatial compatibility between data region and legend reduces response times (RTs) - supported by significant main effects of compatibility on RTs in Experiments 1, 2, and 3 (e.g., Exp 1: $F(1,17)=13.63$, $p=.002$; Exp 2: $F(1,15)=74.7$, $p<.001$; Exp 3: $F(1,17)=58.12$, $p<.001$).
- Spatial incompatibility increases the number of gaze transitions between data region and legend - supported by main effects of compatibility on gaze transitions in all experiments (Exp 1: $F(1,17)=6.05$, $p=.025$; Exp 2: $F(1,15)=31.82$, $p<.001$; Exp 3: $F(1,17)=7.30$, $p=.015$).
- Gaze transitions correlate positively with RTs, indicating that more transitions are associated with slower processing - supported by significant correlations in Experiments 1 (mean $r\approx.55$), 2 (mean $r\approx.64$), and 3 (mean $r\approx.64$).
- Compatibility advantage persists when information must be retrieved from memory - supported by the authors' claim in the general discussion that Experiment 4 replicated the

compatibility effect in a memory-based task (though statistical details are not reported, the authors treat it as evidence).

- Compatibility improves accuracy for bar graphs - supported by a significant main effect of compatibility on accuracy in Experiment 1 ($F(1,17)=5.15$, $p=.037$).
- Compatibility effect is stronger for interaction judgments than for main-effect judgments in simpler graphs - supported by the interaction of compatibility $\times$ effect type on RTs in Experiments 1 and 2 (Exp 1: $F(1,17)=5.79$, $p=.028$; Exp 2: $F(1,15)=43.2$, $p<.001$).

Unsupported or Weakly Supported Claims

- Compatibility universally improves accuracy - not supported for line graphs (Experiment 2 showed no significant accuracy effect, $F(1,15)=2.76$, $p=.118$) and for complex graphs (Experiment 3 showed no accuracy effect, $F<1$).
- Compatibility effects on main-effect RTs are robust across all graph types - unsupported in Experiments 1 and 2 where no RT benefit was found for main effects (Exp 1 $t<1$; Exp 2 $t<1$). Only in Experiment 3 (high visual complexity) did main-effect RTs show a compatibility benefit.
- Compatibility effects are independent of visual complexity - partially unsupported because the effect on main-effect RTs emerged only when visual complexity was increased (Experiment 3), contradicting the claim that compatibility alone drives performance.
- The spatial-compatibility manipulation isolates pure S-S compatibility - weakly supported; the critique notes that legend position, size, and overall layout co-varied with compatibility, potentially confounding the predictor.

Contradictions or Tensions

- The authors claim that spatial compatibility is a major determinant of integration, yet the compatibility effect on RTs is absent for main effects in Experiments 1 and 2 and for accuracy in Experiment 2, suggesting the effect is conditional rather than universal.
- The discussion extends the proximity compatibility principle to relative spatial proximity, but the empirical evidence is limited to a narrow set of graph formats (single bar or line graphs) and does not test other graph types, creating a tension between the broad theoretical generalization and the limited scope of the data.
- Experiment 4 is presented as confirming the compatibility effect in memory-based judgments, but the methods and results lack explicit reporting of the compatibility manipulation and statistical outcomes, creating a mismatch between the claimed evidence and the documented data.

Strengths

- Methodologically rigorous within-subject designs across four experiments, systematically varying spatial compatibility, effect type, and visual complexity.
- Integration of eye-tracking data provides converging evidence: compatible graphs yielded fewer gaze transitions and these transitions correlated positively with response times, supporting the hypothesized integration difficulty.
- Clear theoretical framing linking spatial compatibility literature to the integration phase of the Carpenter & Shah graph comprehension model, and explicit predictions about interaction versus main-effect judgments.
- Demonstration of a compatibility advantage not only in immediate graph judgments (Experiments 1-3) but also when information must be retrieved from memory (Experiment 4), suggesting ecological relevance.
- Comprehensive reporting of effect sizes (ηp^2) for key RT, accuracy, and eye-movement findings, allowing assessment of practical significance.
- Cross-experiment synthesis showing that visual complexity moderates the compatibility effect, thereby refining the theoretical claim and identifying boundary conditions.

Weaknesses and Concerns

- The spatial-compatibility manipulation co-varied with other layout features (legend position, size, and overall separation), confounding pure S-S compatibility with broader visual design differences and limiting causal inference.
- Outcome measures (RT, accuracy, gaze transitions) were aggregated under a single "graph comprehension" construct despite reflecting distinct cognitive processes; this reduces the precision of the test of the integration-specific hypothesis.
- Key compatibility effects were absent for main-effect RTs in Experiments 1 and 2 and for accuracy in Experiment 2, yet the discussion presents compatibility as a general determinant of integration, overstating its universality.
- Experiment 4 lacks a clearly defined compatibility manipulation and provides no statistical results, creating a mismatch between the claimed memory-based replication and the documented evidence.
- Statistical reporting is incomplete: Experiment 3 omits results for the 'match' factor, and several post-hoc tests lack explicit multiple-comparison corrections, raising concerns about Type I error inflation.
- Methodological robustness is weakened by small sample sizes ($N=12-18$), absence of a priori power analyses, and missing details on exclusion criteria, counterbalancing

(Experiments 3-4), and eye-movement coding reliability.

Key References

- **Proctor & Vu (2006)** -- foundational theory
- **Hommel & Prinz (1997)** -- foundational theory
- **Kornblum (1992)** -- foundational theory
- **Carpenter and Shah (1998)** -- primary methodological precedent
- **Feeny, Holo, Liversedge, Findlay, & Metcalf (2000)** -- key empirical precedent
- **Renshaw, Finlay, Tyfa, & Ward (2004)** -- key empirical precedent
- **Gillan, Wickens, Hollands, & Carswell (1998)** -- competing theoretical account
- **Wickens & Carswell (1995)** -- foundational theory
- **Wickens & Hollands (2000)** -- foundational theory
- **Halford, Wilson, & Phillips (1998)** -- key empirical precedent
- **Ratwani, Trafton, & Boehm-Davis (2008)** -- key empirical precedent
- **Kintsch, Welsch, Schmalhofer, & Zimny (1990)** -- alternative theoretical account

Abstract

Preserved from source without summarization.

Abstract A precondition for efficiently understanding and memorizing graphs is the integration of all relevant graph elements and their meaning. In the present study, we analyzed integration processes by manipulating the spatial compatibility between elements in the data region and the legend. In Experiment 1, participants judged whether bar graphs depicting either statistical main effects or interactions correspond to previously presented statements. In Experiments 2 and 3, the same was tested with line graphs of varying complexity. In Experiment 4, participants memorized line graphs for a subsequent validation task. Throughout the experiments, eye movements were recorded. The results indicated that data-legend compatibility reduced the time needed to understand graphs, as well as the time needed to retrieve relevant graph information from memory. These advantages went hand in hand with a decrease of gaze transitions between the data region and the legend, indicating that data-legend compatibility decreases the difficulty of integration processes.

Introduction (Paper-Level)

Research Gap / Motivation

The narrow-sense spatial (S-R and S-S) compatibility has been largely neglected in graph comprehension research, and its specific contribution to integration processes remains unclear.

Theoretical Framework

spatial compatibility influences performance

Compatibility has been subject to a long research tradition (e.g., Proctor & Vu, 2006), and it is widely known that compatibility enhances performance by decreasing the amount of errors and/or response times (RTs; see Hommel & Prinz, 1997; Kornblum, 1992).

- Citations: Proctor & Vu (2006); Hommel & Prinz (1997); Kornblum (1992)

integration processes in graph comprehension

An influential model of graph comprehension comprising three phases was introduced by Carpenter and Shah (1998). An initial pattern recognition phase is devoted to encoding, an interpretation phase involves retrieving qualitative and quantitative meanings, and an integration process relates meanings to semantic referents defined by labels, legends, or titles.

- Citations: Carpenter and Shah (1998)

Specific Hypotheses / Predictions

incompatibility hampers graph comprehension

Thus, the aim of the present study was to test whether spatial incompatibility between the data region and the legend hampers graph comprehension and whether this effect is found across different display types (bar graphs vs. line graphs).

complexity moderates compatibility effect

We systematically varied various forms of complexity to further specify the conditions under which compatibility effects may occur during graph comprehension.

Mechanistic Claims

incompatibility increases gaze transitions

Throughout the experiments, eye movements were recorded to further specify integration processes by deciding whether the expected adverse effects of incompatibility were due to an increased frequency of gaze transitions between the data region and the legend.

compatibility facilitates integration

we suppose that compatibility between data region and legend can play a crucial role for integration processes in graph comprehension.

Constructs of Interest

- **spatial compatibility (data-legend relation)** (*predictor*): the spatial alignment between the data region of a graph and its legend entries; compatible when aligned, incompatible when misaligned.
- **graph comprehension** (*outcome*): performance measured by response times, accuracy, and eye-movement metrics (e.g., number of gaze transitions) during graph tasks.

Rationale for Predictor

Compatibility has been shown to enhance performance across many domains, and prior work suggests that spatial mismatches between legend and data may increase cognitive load during integration.

Rationale for Outcome

Graph comprehension is the central task of interpreting visual data; response time, accuracy, and eye-movement patterns provide sensitive indices of integration efficiency.

Key Terminology

spatial compatibility: the relation of the physical features of stimuli and responses (S-R compatibility) or of features of different elements in a stimulus display (S-S compatibility).

According to a narrower sense, compatibility refers to the relation of the physical features of stimuli and responses (S-R compatibility) or of features of different elements in a stimulus display (S-S compatibility) (Fitts & Simon, 1952; Proctor & Vu, 2006).

integration processes: the phase in graph comprehension that relates meanings to semantic referents defined by labels, legends, or titles.

An integration process relates meanings to semantic referents defined by labels, legends, or titles.

Background Research

- Larkin & Simon (1987) showed that graphs complement verbal information in visual media.
- Carpenter & Shah (1998) introduced a three-phase model of graph comprehension.
- Feeny et al. (2000) manipulated spatial compatibility of legend entries and found eye-movement differences.
- Renshaw et al. (2004) compared optimal versus suboptimal graph layouts and reported distinct eye-movement patterns.
- Ratwani, Traflet, & Boehm-Davis (2008) distinguished visual and cognitive integration processes in graph reading.

Alternative Theories Discussed

surface-information view

- **Description:** Legend inspection yields only surface memory codes that are task-irrelevant, implying low working-memory demands and no adverse effect of incompatibility.
- **Difference from focal theory:** Posits that spatial position of legend entries is irrelevant and should not affect integration, contrasting with the compatibility-based account that predicts performance decrements when spatial relations are mismatched.
- **Citations:** Kintsch, Welsch, Schmalhofer, & Zimny (1990)

Overview of Studies

Four experiments were conducted: Experiments 1-2 examined bar and line graphs with varying data-pattern complexity; Experiment 3 added visual complexity by increasing data quantity; Experiment 4 required memory-based judgments, all while recording eye movements.

Missing or Unclear

- Explicitly named overarching theory (no specific label provided)
- Quantitative effect size predictions
- Formal mediation or moderation models

All Citations Referenced

- **Larkin & Simon (1987)** -- graphs complement verbal information
- **Harris (1999)** -- software for graph visualization
- **Tufte (1983)** -- data-ink ratio
- **Carpenter & Shah (1998)** -- graph comprehension model
- **Carswell, Frankenberger, & Bernhard (1991)** -- design aspects evaluated
- **Fischer (2000)** -- compatibility studies
- **Shah & Carpenter (1995)** -- compatibility studies
- **Siegrist (1996)** -- compatibility studies
- **Spence (1990)** -- compatibility studies
- **Zacks, Levy, Tversky, & Schiano (1998)** -- design aspects
- **Proctor & Vu (2006)** -- compatibility tradition
- **Hommel & Prinz (1997)** -- compatibility enhances performance
- **Kornblum (1992)** -- compatibility enhances performance
- **Sparrow (1989)** -- graph layout compatibility
- **Vessey (1991,1994)** -- graph layout compatibility
- **Washburne (1927)** -- graph layout compatibility
- **Wickens & Andre (1990)** -- graph layout compatibility
- **Wickens & Carswell (1995)** -- graph layout compatibility
- **Meyer (2000)** -- tables for exact data
- **Zacks & Tversky (1999)** -- line graphs for trends
- **Meyer, Shinar, & Leiser (1997)** -- bar graphs for maxima
- **Pinker (1990)** -- expectations based on knowledge
- **Fischer, Dewulf, & Hill (2005)** -- familiar layouts speed comprehension
- **Gattis & Holyoak (1996)** -- familiar layouts speed comprehension
- **Shah (1995)** -- familiar layouts speed comprehension
- **Fitts & Simon (1952)** -- S-R and S-S compatibility
- **Feeny, Holo, Liversedge, Findlay, & Metcalf (2000)** -- spatial compatibility of legend
- **Renshaw, Finlay, Tyfa, & Ward (2004)** -- eye-tracking of optimal vs. suboptimal layouts
- **Carpenter & Shah (1998)** -- integration received little empirical attention

- **Carswell & Wickens (1990)** -- spatial design characteristics
- **Gillan (1995)** -- spatial design characteristics
- **Lohse (1991)** -- spatial design characteristics
- **Bryant & Tversky (1999)** -- integration processes
- **Zacks & Tversky (1999)** -- integration processes
- **Ratwani, Trafton, & Boehm-Davis (2008)** -- visual vs. cognitive integration
- **Peebles (2008)** -- integration determined by graph types
- **Peebles & Cheng (2001,2003)** -- integration determined by user characteristics
- **Gillan, Wickens, Hollands, & Carswell (1998)** -- direct labeling reduces legend difficulty
- **Kosslyn (1994)** -- direct labeling reduces legend difficulty
- **Schmid (1983)** -- direct labeling reduces legend difficulty
- **APA (2006)** -- incompatible legends in guidelines
- **APA (2010)** -- incompatible legends in guidelines
- **Halford, Wilson, & Phillips (1998)** -- interactions more complex than main effects
- **Kintsch, Welsch, Schmalhofer, & Zimny (1990)** -- surface information in text comprehension

Additional Quoted Evidence

Compatibility has been subject to a long research tradition ... and it is widely known that compatibility enhances performance by decreasing the amount of errors and/or response times (RTs).

An integration process relates meanings to semantic referents defined by labels, legends, or titles.

Thus, the aim of the present study was to test whether spatial incompatibility between the data region and the legend hampers graph comprehension.

Experiment 1 and 2 -- Introduction

Section: exp_1_and_2

Rationale for This Experiment

To examine how legend-compatibility affects graph comprehension in bar graphs (Experiment 1) and line graphs (Experiment 2), and to determine whether observed effects are due to graph

type or specific layout decisions.

Relation to Prior Experiments

These are the first experiments reported in the paper; they introduce the core manipulations that later experiments will build upon.

New Elements Introduced

- Bar graphs with legend placed above the data region
- Line graphs with legend presented on the right side
- Separate experimental designs for bar and line graphs to avoid confounding layout differences

Constraints Addressed

- Controls for arbitrary layout decisions that differ between bar and line graphs

Missing or Unclear

- Specific predictions tested in each experiment
- Any theoretical refinements or extensions beyond the paper-level introduction
- Mention of alternative theories not previously discussed

Experiment 1 and 2 -- Methods

Section: exp_1_and_2

Participants

- **N:** 18 (Experiment 1) and 16 (Experiment 2)
- **Demographics:** Experiment 1: 18 students (14 female, 4 male), mean age 26 (range 20-48). Experiment 2: 16 students (14 female, 2 male), mean age 22 (range 20-25). Normal or corrected-to-normal vision.
- **Testing context:** Lab setting, individual testing, single 20-min session per experiment.

Design

Design statement: 2 (compatibility: compatible vs. incompatible) x 2 (effect type: main effect vs. interaction) within-subjects design.

Predictors:

- **compatibility**
 - Levels: compatible, incompatible
 - Manipulation: within-subjects
 - Factor type: nominal
- **effect type**
 - Levels: main effect, interaction
 - Manipulation: within-subjects
 - Factor type: nominal

Counterbalancing: Response key mapping (match vs. mismatch) counterbalanced across participants.

Outcome Measures

- **manual reaction time:** RTs for correct trials measured from graph onset to keypress.
- **accuracy:** Percentage of correct match/mismatch responses.
- **gaze transition frequency:** Number of transitions between data region and legend per trial.

Materials

- **Description:** Bar graphs (Experiment 1) and grayscale line graphs (Experiment 2) with legends; bar graphs had four bars (two gray, two black) representing screen viewing time; line graphs had two uncrossed lines with black/gray markers.
- **Source:** Generated with Microsoft Excel (in-house).
- **Stimulus pool size:** 32 distinct graph figures (templates varied to yield compatible/incompatible relations).
- **Stimuli presented per participant:** 64 trials per participant (each of the 32 graphs paired with a matching and a non-matching statement).
- **Additional details:**
 - Legend size 24°×1° (Exp1) and 6°×4° (Exp2); spatial separation 2° (Exp1) and 4° (Exp2).
 - Statements presented on black background, two text lines, 30° horizontal extent.

Procedure

Ordered steps:

1. Visual instruction displayed.
 2. Four practice trials.
 3. 64 experimental trials presented in random order.
 4. Each trial: statement presented; participant ends viewing by pressing space bar.
 5. Fixation cross (250 ms).
 6. Graph presented; participant responds via mouse buttons (match vs. mismatch).
 7. Blank screen (600 ms) before next trial.
- **Apparatus:** 21-in CRT monitor (100 Hz, 1240×1068 px) at 67 cm; EyeLink II head-mounted eyetracker (500 Hz) with chinrest; German QWERTZ keyboard; mouse for responses.
 - **Experimenter presence:** not reported
 - **Presentation format:** Randomized experimental trials.
 - **Stimulus timing:** Statement self-paced until space bar; fixation 250 ms; blank interval 600 ms.
 - **Session duration:** 20 min per experiment.

Preregistration and Analysis Plan

- **Preregistration:** not reported

Data analysis plan:

- Two-way repeated-measures ANOVAs on RTs and accuracy.
- Post-hoc Bonferroni-corrected t-tests where appropriate.

Author-Provided Rationale for Methodological Choices

- Smaller bars always placed on the left side to avoid introducing an additional variable.
- Gaze analyses focused on transition frequency as the most informative parameter.

Missing or Unclear

- Recruitment source not specified.

- Compensation details not provided.
- Power analysis not reported.
- Explicit exclusion criteria not stated.
- Experimenter presence not described.
- Exact duration of statement presentation not reported (self-paced).

Quoted Evidence from Source

Eighteen students (14 of them female and 4 male) with a mean age of 26 years (range: 20-48) participated in Experiment 1.

Design Compatibility (legend spatially compatible vs. incompatible with data region) and the depicted effect type ... were within-subjects independent variables.

The experiment was run in a single session of 20 min and consisted of a visual instruction followed by 4 practice trials ... and 64 randomized experimental trials.

Experiment 1 and 2 -- Results and Discussion

Section: exp_1_and_2

Statistical Approach

The statistical approach involved ANOVAs and t-tests to analyze the effects of compatibility and depicted effect type on response times, accuracy, and gaze transitions.

Key Findings

RTs were greater for incompatible legends than for compatible legends in Experiment 1

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** compatibility, depicted effect type, RTs

Verbatim statistical report:

RTs were greater for incompatible legends than for compatible legends, $F(1,17) = 13.63$, $p = .002$, $\eta_p^2 = .45$

RTs were greater for the judgment of depicted interactions than main effects in Experiment 1

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** compatibility, depicted effect type, RTs

Verbatim statistical report:

RTs were greater for the judgment of depicted interactions, as compared with depicted main effects, $F(1,17) = 14.38$, $p = .001$, $\eta_p^2 = .46$

Significant interaction of compatibility and depicted effect type on RTs in Experiment 1

- **Test:** ANOVA
- **Effect type:** interaction
- **Variables in model:** compatibility, depicted effect type, RTs

Verbatim statistical report:

There was a significant interaction of compatibility and the depicted effect type, $F(1,17) = 5.79$, $p = .028$, $\eta_p^2 = .25$

No significant RT effect of compatibility for the judgment of main effects in Experiment 1

- **Test:** t-test
- **Effect type:** post-hoc comparison
- **Variables in model:** compatibility, RTs

Verbatim statistical report:

Posthoc t-tests revealed that there was no significant RT effect of compatibility for the judgment of main effects (compatible, $M = 3.33$ s, $SE = 0.36$; incompatible, $M = 3.42$ s, $SE = 0.36$) $t < 1$

Verbatim descriptives:

compatible, $M = 3.33$ s, $SE = 0.36$; incompatible, $M = 3.42$ s, $SE = 0.36$

Significant RT effect of compatibility for the judgment of interactions in Experiment 1

- **Test:** t-test
- **Effect type:** post-hoc comparison
- **Variables in model:** compatibility, RTs

Verbatim statistical report:

there was for interactions (compatible, $M = 3.85$ s, $SE = 0.38$; incompatible, $M = 4.83$ s, $SE = 0.57$), $t(17) = 3.21$, $p = .005$

Verbatim descriptives:

compatible, $M = 3.85$ s, $SE = 0.38$; incompatible, $M = 4.83$ s, $SE = 0.57$

Significant effect of compatibility on accuracy in Experiment 1

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** compatibility, accuracy

Verbatim statistical report:

Accuracy, computed as the percentage of correct responses for each participant, amounted to 83.90% ($SE = 3.10$) for compatible graphs, as compared with 76.40% ($SE = 4.70$) for incompatible graphs, yielding a significant effect of compatibility, $F(1,17) = 5.15$, $p = .037$, $\eta_p^2 = .23$

Verbatim descriptives:

compatible graphs, 83.90% ($SE = 3.10$); incompatible graphs, 76.40% ($SE = 4.70$)

Accuracy was greater for the judgment of depicted main effects than interactions in Experiment 1

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** depicted effect type, accuracy

Verbatim statistical report:

Accuracy was greater for the judgment of depicted main effects (87.1%), as compared with interactions (76.4%), $F(1,17) = 12.94$, $p = .002$, $\eta_p^2 = .43$

Verbatim descriptives:

main effects (87.1%); interactions (76.4%)

No significant interaction of depicted effect type and compatibility on accuracy in Experiment 1

- **Test:** ANOVA
- **Effect type:** interaction
- **Variables in model:** depicted effect type, compatibility, accuracy

Verbatim statistical report:

There was no significant interaction of depicted effect type and compatibility, $F < 1$

Mean number of gaze transitions between data region and legend was higher for incompatible graphs in Experiment 1

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** compatibility, gaze transitions

Verbatim statistical report:

The mean number of gaze transitions between data region and legend was significantly higher during the inspection of incompatible graphs ($M = 2.68$ transitions, $SE = 0.41$), as compared with compatible graphs ($M = 2.39$ transitions, $SE = 0.39$), $F(1,17) = 6.05$, $p = .025$, $\eta_p^2 = .26$

Verbatim descriptives:

incompatible graphs ($M = 2.68$ transitions, $SE = 0.41$); compatible graphs ($M = 2.39$ transitions, $SE = 0.39$)

No main effect of depicted effect type on gaze transitions in Experiment 1

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** depicted effect type, gaze transitions

Verbatim statistical report:

There was no main effect of the depicted effect type, and no significant interaction between depicted effect type and compatibility (both $F_s < 1$)

Gaze transitions from legend to data region occurred more often than from data region to legend in Experiment 1

- **Test:** t-test
- **Effect type:** t-test
- **Variables in model:** gaze transitions

Verbatim statistical report:

gaze transitions from legend to data region occurred more often ($M = 1.37$, $SE = 0.20$) than gaze transitions from data region to legend ($M = 1.17$, $SE = 0.20$), $t(17) = 3.18$, $p = .005$

Verbatim descriptives:

legend to data region ($M = 1.37$, $SE = 0.20$); data region to legend ($M = 1.17$, $SE = 0.20$)

Significant positive correlations between the number of gaze transitions and RTs in Experiment 1

- **Test:** correlation
- **Effect type:** correlation
- **Variables in model:** gaze transitions, RTs

Verbatim statistical report:

As a result, for all but 1 participant, we found significant positive correlations between the number of gaze transitions and RTs [mean $r = .55$, $CI_{.95} = .55 < r(63) < .67$]

Verbatim descriptives:

mean $r = .55$, $CI_{.95} = .55 < r(63) < .67$

RTs were greater for incompatible legends than for compatible legends in Experiment 2

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** compatibility, RTs

Verbatim statistical report:

RTs were greater for incompatible legends than for compatible legends, $F(1,15) = 74.7$, $p < .001$, $\eta^2 = .83$

No significant main effect of depicted effect type on RTs in Experiment 2

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** depicted effect type, RTs

Verbatim statistical report:

There was no significant main effect of depicted effect type, $F(1, 15) = 2.9$, $p = .110$

Significant interaction of depicted effect type and compatibility on RTs in Experiment 2

- **Test:** ANOVA
- **Effect type:** interaction
- **Variables in model:** depicted effect type, compatibility, RTs

Verbatim statistical report:

there was a significant interaction of depicted effect type and compatibility, $F(1, 15) = 43.2$, $p < .001$, $\eta_p^2 = .74$

No compatibility effect for depicted main effects on RTs in Experiment 2

- **Test:** t-test
- **Effect type:** post-hoc comparison
- **Variables in model:** compatibility, RTs

Verbatim statistical report:

Posthoc t-tests revealed that there was no compatibility effect for depicted main effects (compatible, 3.70 s, SE = 0.28; incompatible, 3.65 s, SE = 0.30), $t < 1$

Verbatim descriptives:

compatible, 3.70 s, SE = 0.28; incompatible, 3.65 s, SE = 0.30

Significant compatibility effect for depicted interactions on RTs in Experiment 2

- **Test:** t-test
- **Effect type:** post-hoc comparison
- **Variables in model:** compatibility, RTs

Verbatim statistical report:

there was for interactions (compatible, $M = 3.49$ s, $SE = 0.29$; incompatible, $M = 4.67$ s, $SE = 0.32$), $t(15) = 9.31$, $p < .001$

Verbatim descriptives:

compatible, $M = 3.49$ s, $SE = 0.29$; incompatible, $M = 4.67$ s, $SE = 0.32$

No significant effect of compatibility on accuracy in Experiment 2

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** compatibility, accuracy

Verbatim statistical report:

Accuracy amounted to 95.65% ($SE = 1.00$) for compatible graphs, as compared with 93.55% ($SE = 1.50$) for incompatible graphs. However, this tendency yielded no significant effect of compatibility, $F(1,15) = 2.76$, $p = .118$

Verbatim descriptives:

compatible graphs, 95.65% ($SE = 1.00$); incompatible graphs, 93.55% ($SE = 1.50$)

No significant effect of depicted effect type on accuracy in Experiment 2

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** depicted effect type, accuracy

Verbatim statistical report:

There was neither a significant effect of depicted effect type, $F < 1$

No significant interaction of depicted effect type and compatibility on accuracy in Experiment 2

- **Test:** ANOVA
- **Effect type:** interaction

- **Variables in model:** depicted effect type, compatibility, accuracy

Verbatim statistical report:

nor a significant interaction, $F < 1$

Gaze transitions between data region and legend were more frequent in incompatible graphs in Experiment 2

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** compatibility, gaze transitions

Verbatim statistical report:

Gaze transitions between data region and legend were more frequent in incompatible ($M = 4.37$) than in compatible ($M = 3.63$) graphs, $F(1,15) = 31.82$, $p < .001$, $\eta_p^2 = .68$

Verbatim descriptives:

incompatible ($M = 4.37$); compatible ($M = 3.63$)

No main effect of depicted effect type on gaze transitions in Experiment 2

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** depicted effect type, gaze transitions

Verbatim statistical report:

There was no main effect of the depicted effect type, $F(1,15) = 1.72$, $p = .210$

Significant interaction between depicted effect type and compatibility on gaze transitions in Experiment 2

- **Test:** ANOVA
- **Effect type:** interaction
- **Variables in model:** depicted effect type, compatibility, gaze transitions

Verbatim statistical report:

but there was a significant interaction between depicted effect type and compatibility, $F(1,15) = 6.63$, $p = .021$, $\eta_p^2 = .31$

Significant difference between compatible and incompatible legends for main effects on gaze transitions in Experiment 2

- **Test:** t-test
- **Effect type:** post-hoc comparison
- **Variables in model:** compatibility, gaze transitions

Verbatim statistical report:

Posthoc t-tests revealed that there was a significant difference between compatible and incompatible legends when participants had to judge a main effect ($M = 3.70$ gaze transitions, $SE = 0.28$, vs. $M = 4.04$ gaze transitions, $SE = 0.30$), $t(15) = 1.95$, $p = .035$

Verbatim descriptives:

compatible ($M = 3.70$ gaze transitions, $SE = 0.28$); incompatible ($M = 4.04$ gaze transitions, $SE = 0.30$)

Significant difference between compatible and incompatible legends for interactions on gaze transitions in Experiment 2

- **Test:** t-test
- **Effect type:** post-hoc comparison
- **Variables in model:** compatibility, gaze transitions

Verbatim statistical report:

as well as for the interpretation of interactions ($M = 3.56$ gaze transitions, $SE = 0.32$, vs. $M = 4.69$ gaze transitions, $SE = 0.42$), $t(15) = 4.95$, $p < .001$

Verbatim descriptives:

compatible ($M = 3.56$ gaze transitions, $SE = 0.32$); incompatible ($M = 4.69$ gaze transitions, $SE = 0.42$)

Gaze transitions from legend to data region occurred more often than from data region to legend in Experiment 2

- **Test:** t-test
- **Effect type:** t-test
- **Variables in model:** gaze transitions

Verbatim statistical report:

Overall, gaze transitions from legend to data region occurred more often ($M = 2.08$, $SE = 0.14$) than gaze transitions from data region to legend ($M = 1.92$, $SE = 0.16$), $t(15) = 2.56$, $p = .022$

Verbatim descriptives:

legend to data region ($M = 2.08$, $SE = 0.14$); data region to legend ($M = 1.92$, $SE = 0.16$)

Significant positive correlations between the number of gaze transitions and RTs in Experiment 2

- **Test:** correlation
- **Effect type:** correlation
- **Variables in model:** gaze transitions, RTs

Verbatim statistical report:

For all participants, we found significant positive correlations between the number of gaze transitions and RTs (mean $r = .64$, $CI_{.95} = .51 < r(63) < .74$)

Verbatim descriptives:

mean $r = .64$, $CI_{.95} = .51 < r(63) < .74$

Accuracy was greater for line graphs than for bar graphs for depicted main effects

- **Test:** t-test
- **Effect type:** t-test
- **Variables in model:** graph type, depicted effect type, accuracy

Verbatim statistical report:

A between-experiments comparison revealed that accuracy was greater for line graphs than for bar graphs for depicted main effects, $t(32) = 2.5$, $p = .02$

Accuracy was greater for line graphs than for bar graphs for depicted interactions

- **Test:** t-test
- **Effect type:** t-test
- **Variables in model:** graph type, depicted effect type, accuracy

Verbatim statistical report:

and for depicted interactions, $t(32) = 4.3$, $p < .001$

RTs did not differ significantly between line and bar graphs for main effects

- **Test:** t-test
- **Effect type:** t-test
- **Variables in model:** graph type, depicted effect type, RTs

Verbatim statistical report:

RTs did not differ significantly (both $t_s < 1$)

RTs did not differ significantly between line and bar graphs for interactions

- **Test:** t-test
- **Effect type:** t-test
- **Variables in model:** graph type, depicted effect type, RTs

Verbatim statistical report:

RTs did not differ significantly (both $t_s < 1$)

Overall, gaze transitions were more frequent in Experiment 1 (bar graphs)

- **Test:** t-test
- **Effect type:** t-test
- **Variables in model:** graph type, gaze transitions

Verbatim statistical report:

Overall, gaze transitions were more frequent in Experiment 1 (bar graphs), $t(32) = 5.3$, $p < .001$

Authors' Interpretations

compatible graphs judged more quickly

The results of Experiments 1 and 2 clearly revealed that compatible graphs were judged more quickly, more accurately (Experiment 1), and with fewer gaze transitions between data region and legend, as compared with incompatible graphs.

compatible graphs judged more accurately

The results of Experiments 1 and 2 clearly revealed that compatible graphs were judged more quickly, more accurately (Experiment 1), and with fewer gaze transitions between data region and legend, as compared with incompatible graphs.

compatible graphs had fewer gaze transitions

The results of Experiments 1 and 2 clearly revealed that compatible graphs were judged more quickly, more accurately (Experiment 1), and with fewer gaze transitions between data region and legend, as compared with incompatible graphs.

legend incompatibility increased integration difficulty

When gaze transitions represent a marker for integration processes (Carpenter & Shah,1998), this indicates that legend incompatibility indeed increased the difficulty of integrating relevant information from the data and the legend region, finally slowing down RTs.

- Citations: Carpenter & Shah (1998)

RTs positively correlated with gaze transitions

This claim is further corroborated by the observation that RTs were positively correlated with the amount of gaze transitions.

compatibility effect increased for interactions

A closer inspection of the RT data revealed that the compatibility effect was increased (and sometimes only present) for the judgment of interactions, suggesting that a certain threshold of data pattern complexity and a respective increase of cognitive processing demands (Halford et al.,1998) needs to be crossed to achieve an effect of compatibility on performance in terms of RTs and/or accuracy.

- Citations: Halford et al. (1998)

bar graphs produce too much visual clutter

Probably, bar graphs generally produce too much visual clutter, and their use should be more limited than was previously assumed.

decoding of bar graphs was demanding

The overall smaller number of gaze transitions in Experiment 1 may indicate that decoding of bar graphs was probably so demanding that participants tended to avoid rechecking the legend before responding, leading to poor overall performance.

participants less sensitive to legends in bar graphs

Alternatively, participants might have been less sensitive to the trial-by-trial change in the legends because of a lower salience of legends in bar graphs, due to the overall higher amount of "ink."

Authors' Stated Links to Their Own Predictions

compatibility effect increased for interactions

A closer inspection of the RT data revealed that the compatibility effect was increased (and sometimes only present) for the judgment of interactions, suggesting that a certain threshold of data pattern complexity and a respective increase of cognitive processing demands (Halford et al.,1998) needs to be crossed to achieve an effect of compatibility on performance in terms of RTs and/or accuracy.

- Citations: Halford et al. (1998)

Alternative Interpretations Discussed by Authors

- decoding of bar graphs was demanding
- participants less sensitive to legends in bar graphs

Preview of Next Experiment

To further examine the influence of complexity, we conducted Experiment 3, in which more complex line graphs (consisting of more lines and more legend entries) were used.

Missing or Unclear (Non-Statistical)

- No explicit mention of reliability estimates for the measures used.
- No detailed description of the coding procedures for gaze transitions.

Citations Referenced

- **Carpenter & Shah (1998)** -- gaze transitions as a marker for integration processes
- **Halford et al. (1998)** -- cognitive processing demands
- **Gillan et al. (1998)** -- previous guidelines for bar graphs

Experiment 3 -- Introduction

Rationale for This Experiment

To examine whether increased visual complexity of graphs alters the strength or presence of spatial compatibility effects, particularly whether compatibility influences main effects as well as interactions.

Relation to Prior Experiments

Extends earlier experiments that used simpler graphs with fewer legend entries, testing the boundary condition of visual complexity on the compatibility effect.

Specific Predictions

compatibility effects on main effects in complex graphs

If this enhances the potential for compatibility effects, one might expect RT effects not only for depicted interactions, but also for main effects.

Theoretical Refinements (relative to paper-level intro)

modulation of compatibility by visual complexity

In Experiment 3, we explored to what extent the compatibility effect is modulated in graphs with greater visual complexity, where it is difficult to avoid compatibility issues by labeling lines directly without causing visual cluttering of textual elements.

New Elements Introduced

- graphs with greater visual complexity
- use of line graphs instead of bar graphs
- avoidance of direct line labeling to prevent textual clutter

Constraints Addressed

- boundary condition of visual complexity on compatibility effects
- potential limitation of previous experiments regarding low-complexity stimuli

Experiment 3 -- Methods

Section: exp_3

Participants

- **N:** 18 (16 female, 2 male)
- **Demographics:** Mean age 21 years (range 19-27), right-handed, normal or corrected-to-normal vision
- **Recruitment:** RWTH Aachen University students

Design

Design statement: 2 (spatial compatibility: compatible vs incompatible) x 2 (statement type: main effect vs interaction) x 2 (match: matched vs mismatched) within-subjects design.

Predictors:

- **spatial compatibility**
 - Levels: compatible, incompatible
 - Manipulation: within-subjects
 - Factor type: nominal
- **statement type**
 - Levels: main effect, interaction
 - Manipulation: within-subjects
 - Factor type: nominal
- **match**
 - Levels: matched, mismatched
 - Manipulation: within-subjects
 - Factor type: nominal

Counterbalancing: not reported

Outcome Measures

None extracted

Materials

- **Description:** Graphs depicting popularity of six school subjects for 6th and 9th graders; markers varied in shape (triangle, square, circle) and color (black, white); legends with six

lines.

- **Source:** created for the experiment
- **Stimulus pool size:** 8 different graphs
- **Stimuli presented per participant:** 16 graphs per participant
- **Additional details:**
 - steeper lines randomized across graphs
 - each graph paired with spatially compatible and incompatible legends
 - each graph paired with statements referring to main effects and interactions

Procedure

- **Experimenter presence:** not reported
- **Presentation format:** not reported
- **Stimulus timing:** not reported
- **Session duration:** not reported

Preregistration and Analysis Plan

- **Preregistration:** not reported

Missing or Unclear

- compensation not specified
- exclusion criteria not stated
- testing context (lab/online, individual/group) not reported
- apparatus details (hardware/software) not reported
- detailed procedural steps not described
- stimulus timing (duration, ISI) not reported
- outcome measures not identified
- preregistration status not indicated
- data analysis plan not provided
- power analysis not mentioned
- counterbalancing scheme not described

Quoted Evidence from Source

Eighteen new right-handed students at RWTH Aachen University (16 of them female and 2 male) with a mean age of 21 years (range: 19-27) participated in the experiment.

We designed 8 different graphs with randomized positions for the steeper lines, the specific markers, and the related school subjects.

Each of the resulting 16 graphs was coupled with both a statement ... resulting in 64 trials.

Experiment 3 -- Results and Discussion

Section: exp_3

Key Findings

RTs were greater for incompatible legends than for compatible legends

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** compatibility, effect type, RTs

Verbatim statistical report:

RTs were greater for incompatible legends ($M = 6.45$ s) than for compatible legends ($M = 5.70$ s), $F(1,17) = 58.12$, $p < .001$, $\eta_p^2 = .77$

Verbatim descriptives:

RTs were greater for incompatible legends ($M = 6.45$ s) than for compatible legends ($M = 5.70$ s)

No main effect of effect type on RTs

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** effect type, RTs

Verbatim statistical report:

There was no main effect of effect type, $F(1,17) = 2.98$, $p > .10$

No interaction of effect type and compatibility on RTs

- **Test:** ANOVA
- **Effect type:** interaction
- **Variables in model:** effect type, compatibility, RTs

Verbatim statistical report:

| no interaction of effect type and compatibility, $F(1,17) = 1.74$, $p > .10$

Significant compatibility effect for depicted main effects on RTs

- **Test:** t-test
- **Effect type:** post-hoc comparison
- **Variables in model:** compatibility, RTs for main effects

Verbatim statistical report:

| Posthoc t-tests revealed a significant compatibility effect for depicted main effects, $t(17) = 4.11$, $p = .001$

Significant compatibility effect for interactions on RTs

- **Test:** t-test
- **Effect type:** post-hoc comparison
- **Variables in model:** compatibility, RTs for interactions

Verbatim statistical report:

| Posthoc t-tests revealed a significant compatibility effect for interactions, $t(17) = 7.02$, $p < .001$

No effect of compatibility on accuracy

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** compatibility, accuracy

Verbatim statistical report:

| Accuracy amounted to 95.5% (SE = 0.90) for compatible graphs and 94.5% (SE = 1.10) for incompatible graphs, $F < 1$

Verbatim descriptives:

Accuracy amounted to 95.5% (SE = 0.90) for compatible graphs and 94.5% (SE = 1.10) for incompatible graphs

No main effect of effect type on accuracy

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** effect type, accuracy

Verbatim statistical report:

Accuracy was similar for the judgment of main effects ($M = 96.3\%$, $SE = 1.20$) and for interactions ($M = 96.0\%$, $SE = 0.90$), $F(1,17) = 2.22$, $p > .10$

Verbatim descriptives:

Accuracy was similar for the judgment of main effects ($M = 96.3\%$, $SE = 1.20$) and for interactions ($M = 96.0\%$, $SE = 0.90$)

No interaction of effect type and compatibility on accuracy

- **Test:** ANOVA
- **Effect type:** interaction
- **Variables in model:** effect type, compatibility, accuracy

Verbatim statistical report:

There was no significant interaction of effect type and compatibility, $F(1,17) = 1.89$, $p > .10$

Gaze transitions were more frequent in incompatible graphs

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** compatibility, gaze transitions

Verbatim statistical report:

Gaze transitions were significantly more frequent in incompatible ($M = 4.41$) than in compatible ($M = 4.13$) graphs, $F(1,17) = 7.30$, $p = .015$, $\eta_p^2 = .30$

Verbatim descriptives:

Gaze transitions were significantly more frequent in incompatible ($M = 4.41$) than in compatible ($M = 4.13$) graphs

No main effect of effect type on gaze transitions

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** effect type, gaze transitions

Verbatim statistical report:

There was no main effect of the effect type, $F(1,17) = 1.29$, $p > .10$

No interaction between effect type and compatibility on gaze transitions

- **Test:** ANOVA
- **Effect type:** interaction
- **Variables in model:** effect type, compatibility, gaze transitions

Verbatim statistical report:

no significant interaction between effect type and compatibility, $F < 1$

More gaze transitions from legend to data region than vice versa

- **Test:** t-test
- **Effect type:** t-test
- **Variables in model:** direction of gaze transition, frequency

Verbatim statistical report:

Overall, gaze transitions from legend to data region occurred more often ($M = 2.36$, $SE = 0.12$) than gaze transitions from data region to legend ($M = 1.90$, $SE = 0.09$), $t(17) = 7.21$, $p < .001$

Verbatim descriptives:

Overall, gaze transitions from legend to data region occurred more often ($M = 2.36$, $SE = 0.12$) than gaze transitions from data region to legend ($M = 1.90$, $SE = 0.09$)

High percentage of trials with legend fixated before data region

- **Test:** descriptive statistics
- **Effect type:** descriptive

- **Variables in model:** fixation order, percentage

Verbatim statistical report:

The mean percentage of trials on which the legend was fixated before the data region across participants amounted to 94.3% (SE = 1.47)

Verbatim descriptives:

The mean percentage of trials on which the legend was fixated before the data region across participants amounted to 94.3% (SE = 1.47)

Significant correlation between gaze transitions and RTs

- **Test:** correlation
- **Effect type:** correlation
- **Variables in model:** gaze transitions, RTs

Verbatim statistical report:

For all participants, we found significant correlations between the amount of gaze transitions and RTs (mean $r = .64$, $CI_{.95} = .55 < r(63) < .73$)

Authors' Interpretations

generalization of compatibility effect

Experiment 3 demonstrated that the compatibility effect on RTs and gaze transitions generalizes to more visually complex graphs.

role of visual complexity

Interestingly, we now found a significant compatibility effect on RTs, not only for depicted interactions, but also for main effects, suggesting that visual complexity plays a major role.

processing demands

Probably, the additional task of selecting the relevant pieces of data from the overwhelming amount of information in the data and legend regions increased overall processing demands, subsequently increasing adverse effects of incompatibility.

influence of relevant legend entries

The overall number of gaze transitions was in a similar range as in Experiment 2, suggesting that in complex graphs, the number of relevant legend entries is more

| influential for legend revisitations than is their overall number.

Experiment 4 -- Introduction

Section: exp_4

Rationale for This Experiment

To determine if the spatial compatibility effect observed with visually present graphs also occurs when graph information must be retrieved from memory, extending the findings to more ecologically realistic situations.

Relation to Prior Experiments

Experiment 4 builds on Experiment 2 by using the same graphs and statements but reversing their within-trial presentation order and requiring participants to judge the graph from memory rather than from a currently displayed stimulus.

Theoretical Refinements (relative to paper-level intro)

compatibility effect with memory retrieval

| we asked whether the compatibility effect transfers to a situation where relevant graph information is no longer visually present and, thus, needs to be retrieved from memory.

New Elements Introduced

- Judgment of graphs from memory instead of direct visual inspection
- Reversed within-trial presentation order of graphs and statements
- Limited exposure time for the initial graph presentation

Constraints Addressed

- Tests the boundary condition of visual presence of graph information
- Addresses whether compatibility effects persist when information must be retrieved from memory

Missing or Unclear

- Specific predictions about the direction or magnitude of the compatibility effect in the memory condition are not explicitly stated.

Experiment 4 -- Methods

Section: exp_4

Participants

- **N:** 12 (10 female, 2 male)
- **Demographics:** right-handed students, mean age 22 years (range 19-27)
- **Recruitment:** RWTH Aachen University

Design

Counterbalancing: not reported

Outcome Measures

- **reaction time:** time from statement onset to participant's response (statement verification time)

Materials

- **Description:** line graphs and corresponding statements
- **Source:** same as Experiment 2 (not further specified)
- **Additional details:**
 - fixation cross (250 ms) before graph
 - graph displayed for fixed 5 s

Procedure

Ordered steps:

1. Fixation cross presented for 250 ms
2. Line graph presented for 5 s
3. Statement displayed

4. Participant judges as quickly and accurately as possible whether it corresponds to the graph
 5. Fixation cross of next trial appears
- **Apparatus:** not reported
 - **Experimenter presence:** not reported
 - **Presentation format:** not reported
 - **Stimulus timing:** graph fixed 5 s; statement self-paced

Preregistration and Analysis Plan

- **Preregistration:** not reported

Author-Provided Rationale for Methodological Choices

- The 5-s interval was based on the results from Experiment 2, where mean RTs did not exceed 5 s in any condition.

Missing or Unclear

- full experimental design details
- predictor variables and levels
- counterbalancing scheme
- apparatus specifications
- experimenter presence
- compensation details
- exclusion criteria
- testing context (individual/group, lab/online)
- number of stimuli in pool and presented per participant
- preregistration information
- data analysis plan
- power analysis

Quoted Evidence from Source

Twelve new right-handed students at RWTH Aachen University (10 of them female and 2 male) with a mean age of 22 years (range: 19-27).

the line graph was presented first for a fixed time interval of 5 s, preceded by the fixation cross (250 ms).

participants judged as quickly and accurately as possible whether it corresponded to the previous graph.

Experiment 4 -- Results and Discussion

Not available for this paper.

General Discussion

Section: paper-level

Summary of Findings

The study found that spatial incompatibility between legend and data regions slows response times, increases errors, and raises gaze transitions, especially under higher data-pattern complexity; the effect also persists when information must be retrieved from memory, indicating that compatibility influences integration processes via working-memory demands.

Theoretical Conclusions

working memory interference mechanism

the increase of revisitations to the legend in incompatible graphs may suggest that the adverse effects of incompatibility were due mainly to interference or capacity limits in working memory, and not only to a more time-consuming process of memory retrieval of legend information during the fixation of the data region

expectation of spatial layout

participants did not relate the verbal label solely to the marker information, but also, at least on some trials, to its (task-irrelevant) spatial position. Probably, this task-irrelevant information automatically generates the expectation that the spatial layout in the data region is designed accordingly

spatial compatibility as determinant of integration

On the basis of the present study, we demonstrated that spatial compatibility in combination with task complexity is a major determinant of this process

Cross-Experiment Integration

complexity moderates compatibility effect

Since in Experiments 1 and 2, the compatibility effect on RTs occurred only for the judgment of interactions, and not for main effects... In Experiment 3, the compatibility effect was also present for the judgment of depicted main effects, showing that visual complexity effectively modulated the compatibility effect, too

memory-based decisions replicate effect

Experiment 4 extended the theoretical significance of this compatibility effect by demonstrating that incompatible graphs also affected information retrieval from memory in the absence of direct visual stimulation

Results Authors State Support Their Theory

proximity compatibility principle extension

Our present results bear interesting implications for the proximity compatibility principle, since they demonstrate that display proximity can also take the form of relative spatial proximity (referring to the relative position of elements in the legend and data region), thus extending the original claim that (absolute) spatial proximity plays a crucial role

- Citations: Wickens & Carswell (1995); Wickens & Hollands (2000)

legend-indicator ordering prediction

More specifically, Gillan et al. (1998) explicitly predicted that the ordering of legend symbols should match the order of indicators in the data region, a claim that is directly supported by our data

- Citations: Gillan et al. (1998)

Results Authors State Challenge a Theory

limits of proximity compatibility principle

Thus, the proximity compatibility principle does not, per se, hold in any given situation, but only in tasks of sufficient (data pattern) complexity that a certain amount of cognitive load is induced

Alternative Accounts Addressed by Authors

optimal scan path assumption (*ad-hoc alternative*)

- **Description:** the assumption that an optimal scan path includes only necessary transitions
- **How authors addressed it:** acknowledge and argue against, noting that additional transitions may arise from working-memory limitations or strategic factors
- **Citations:** Peebles & Cheng (2002); Peebles & Cheng (2003)

Implications for the Field

design recommendation for compatible legends

Here, we present the first empirical evidence for the advice to use compatible legends, especially when readers are supposed to extract interactions

legend ordering for line graphs

legend entries in line graphs (with legends on the right side) should be ordered in the same way as the rightmost endpoints of the lines

- Citations: Parkin (1983)

Applied Implications Stated by Authors

- Use compatible legends, particularly when readers need to extract interactions, as incompatibility impairs performance even in memory-based tasks.

Limitations Stated by Authors

- Only single graphs were examined per trial, limiting generalization to multi-graph contexts.
- The experiments always presented an unambiguous compatible legend arrangement, which may not be possible for certain graph types (e.g., disordinal interactions with line crossings).

Future Directions Proposed by Authors

- Investigate how consistent spatial arrangements across multiple graphs affect compatibility effects.
- Examine compatibility effects in graphs where a compatible legend ordering is not feasible.

Citations Referenced

- Wickens & Carswell (1995) --
- Wickens & Hollands (2000) --
- Gillan et al. (1998) --
- Halford et al. (1998) --
- Ratwani et al. (2008) --
- Carpenter & Shah (1998) --
- Peebles & Cheng (2002) --
- Peebles & Cheng (2003) --
- Parkin (1983) -- in Pinker, 1990
- Pinker (1990) --

Critique

Paper:

huestegge-and-philipp-2011-effects-of-spatial-compatibility-on-integration-processes-in-graph-comprehension

Overall Assessment

The paper's methodological rigor is compromised by several high-severity issues, notably the lack of a priori power analysis and small, under-reported sample sizes, which undermine statistical power and generalizability. While the theoretical framework is articulated, the alignment between the spatial-compatibility manipulation and the underlying theory is weak, and key predictions are not empirically contrasted with competing accounts. Evidentiary support for the main claims is limited by incomplete reporting of statistical results, ambiguous outcome measures, and insufficient control of alternative explanations. Consequently, the study falls short of the standards expected for robust scholarly contribution.

1. Design-Theory Alignment

- **The spatial compatibility manipulation also changes overall visual layout (e.g., legend position and distance), so the predictor may not isolate pure S-S compatibility.**
- Severity: **moderate**
- Justification: Spatial compatibility is defined as the alignment between data region and legend entries, but the experimental manipulations simultaneously alter legend proximity,

size, and overall layout. This conflates the intended construct with other visual factors, making it unclear whether observed effects are due solely to compatibility or to broader layout differences.

Evidence:

- methods/exp_1 -- field materials: Bar graphs had legend placed above the data region; line graphs had legend on the right side, with different spatial separations (2° vs 4°).
- methods/exp_2 -- field materials: Legend size and spatial separation differed between bar and line graphs, potentially introducing confounds beyond compatibility.

The outcome measure of "graph comprehension" combines response time, accuracy, and gaze transitions, which may reflect distinct cognitive processes beyond integration.

- Severity: **moderate**
- Justification: By aggregating RTs, accuracy, and eye-movement metrics under a single construct, the study assumes they all index the same theoretical mechanism (integration). However, each metric can be influenced by separate factors (e.g., motor execution, decision criteria, visual scanning), potentially diluting the test of the proposed integration mechanism.

Evidence:

- methods/exp_1 -- field name: Manual reaction time, accuracy, and number of gaze transitions were all treated as indices of comprehension.
- results_and_discussion/exp_1 -- field topic_label: Authors link gaze transitions to integration difficulty, yet transitions could also be driven by visual search strategies unrelated to integration.

In Experiment 4 the predictor variable (spatial compatibility) is not explicitly operationalized, and the design lacks a clear manipulation of compatibility, undermining alignment with the theory.

- Severity: **minor**
- Justification: Without a clearly defined compatibility manipulation, it is impossible to assess whether the memory-based task truly tests the same spatial compatibility mechanism as earlier experiments, weakening the theoretical linkage.
- Evidence:
 - methods/exp_4 -- field design_statement: Design statement is null and predictors are not listed, despite the study's aim to test compatibility effects in a memory-based task.

- methods/exp_4 -- field description: Materials are described only as "line graphs and corresponding statements" without specifying compatible vs. incompatible legend arrangements.

2. Evidence-Theory Alignment

- **The authors claim that spatial compatibility is a major determinant of graph integration, yet the key compatibility effect on response times was absent for main effects in Experiments 1, 2, and 3 and for accuracy in Experiment 2, which the discussion downplays.**

- Severity: **moderate**

- Justification: The theoretical claim that compatibility drives integration is presented as a general principle, yet the empirical pattern shows that the effect is limited to interaction judgments and does not consistently affect accuracy. By not fully integrating these null results, the discussion overstates the universality of the compatibility effect.

Evidence:

- results_and_discussion/exp_1 -- field key_findings: No significant RT effect of compatibility for the judgment of main effects ($t < 1$) and no significant interaction on accuracy ($F < 1$).
- results_and_discussion/exp_2 -- field key_findings: No compatibility effect for depicted main effects on RTs ($t < 1$) and no significant effect of compatibility on accuracy ($F = 2.76, p = .118$).
- results_and_discussion/exp_3 -- field key_findings: No interaction of effect type and compatibility on RTs ($F(1,17) = 1.74, p > .10$) and no main effect of compatibility on accuracy ($F < 1$).

The paper extends the proximity compatibility principle to "relative spatial proximity" and recommends design changes (e.g., ordering legend entries to match line endpoints) despite only testing a narrow set of graph types and layouts.

- Severity: **moderate**

- Justification: The empirical evidence is confined to specific graph configurations (single bar or line graphs with a particular legend placement). Generalizing to a broader design recommendation exceeds the scope of the data, risking overinterpretation of the theory's applicability.

Evidence:

- `general_discussion -- field results_stated_to_support_theory`: Authors claim their results extend the proximity compatibility principle to relative spatial proximity and support Gillan et al.'s legend-indicator ordering prediction.
- `abstract -- field verbatim_quote`: Only bar graphs, simple line graphs, and one memory-based line graph were examined; no other graph formats or multi-graph contexts were tested.

The authors interpret increased gaze transitions as direct evidence for working-memory interference, yet the eye-movement metric could also reflect strategic scanning differences that are not uniquely tied to memory load.

- Severity: **minor**
- Justification: While the correlation between gaze transitions and RTs is reported, the authors do not rule out alternative accounts (e.g., differing visual search strategies). Presenting the metric as definitive evidence for a specific cognitive mechanism overstates the inferential power of the data.
- Evidence:
- `results_and_discussion/exp_1 -- field authors_interpretations`: Interpretation that gaze transitions represent integration difficulty and thus working-memory interference.
- `general_discussion -- field theoretical_conclusions`: Claim that increased revisitations suggest interference or capacity limits in working memory rather than alternative explanations.

4. Alternative Accounts

- **The paper introduces the "surface-information view" as a competing account but never empirically tests or discusses its predictions against the observed compatibility effects.**
- Severity: **moderate**
- Justification: Introducing a plausible competing theory without any empirical evaluation leaves the focal compatibility account under-specified and may overstate its explanatory power. The authors should have either tested predictions derived from the surface-information view (e.g., no effect of spatial layout) or provided a stronger theoretical argument for dismissing it. The omission limits the paper's ability to rule out alternative explanations for the observed performance and eye-movement differences.
- Evidence:

- introduction -- field alternative_theories: The surface-information view is listed as an alternative theory in the paper-level introduction.
- results_and_discussion/exp_1_and_2 -- field alternative_interpretations_discussed: Only alternative interpretations related to bar-graph decoding and legend salience are discussed; the surface-information view is absent.
- results_and_discussion/exp_3 -- field alternative_interpretations_discussed: No mention of the surface-information view despite its earlier introduction.
- results_and_discussion/exp_4 -- field alternative_interpretations_discussed: Again, the surface-information view is not addressed.

5. Statistical Completeness

- **Experiment 3 omitted reporting of any main effects or interactions involving the 'match' factor despite a $2 \times 2 \times 2$ within-subjects design.**
- Severity: **moderate**
- Justification: The methods specify a three-factor design, yet the results only address compatibility and statement type. Standard reporting would include the main effect of match and any two-way or three-way interactions involving match. Their absence constitutes an incomplete statistical report.

Evidence:

- methods/exp_3 -- field design_statement: 2 (spatial compatibility) x 2 (statement type) x 2 (match) within-subjects design.
- results_and_discussion/exp_3 -- field key_findings: All reported ANOVA effects involve only compatibility and effect type; no mention of main effect of match or any interactions with match.

Experiment 4 provides no statistical results despite having a methods section, leaving the analysis of the compatibility effect on memory-based judgments unreported.

- Severity: **moderate**
- Justification: Given the experimental rationale, readers expect statistical tests of compatibility (e.g., main effects, interactions). The complete absence of any reported analyses prevents assessment of whether the hypothesized effects were examined, indicating a gap in statistical completeness.

- Evidence:
- methods/exp_4 -- field design: Predictors and design statement are missing, but the experiment is described as testing compatibility effects with memory retrieval.
- results_and_discussion/exp_4 -- field key_findings: No results entries are present for Experiment 4.

6. Methodological Concerns

- **No a priori power analysis was reported for any of the four experiments despite modest sample sizes**
- Severity: **high**
- Justification: The methods sections for all experiments explicitly note that a power analysis was not reported. Given the small per-experiment Ns (18, 16, 18, 12), the lack of a priori sensitivity analysis raises concerns about whether the studies were adequately powered to detect the reported effects.

Evidence:

- methods/exp_1 -- field power_analysis: null (power analysis not reported)
- methods/exp_2 -- field power_analysis: null (power analysis not reported)
- methods/exp_3 -- field power_analysis: null (power analysis not reported)
- methods/exp_4 -- field power_analysis: null (power analysis not reported)

Sample sizes are small (N=12-18 per experiment) and per-condition Ns are not reported, limiting statistical power and generalizability

- Severity: **moderate**
- Justification: Each experiment reports only a single group of participants with total Ns ranging from 12 to 18, and does not provide the number of trials per condition. Such limited sample sizes increase the risk of Type II errors and reduce confidence in the robustness of the observed effects.

Evidence:

- methods/exp_1 -- field participants: 18 participants (Experiment 1) and 16 participants (Experiment 2)
- methods/exp_3 -- field participants: 18 participants
- methods/exp_4 -- field participants: 12 participants

Exclusion criteria are not specified for any experiment, making it unclear how data were screened

- Severity: **moderate**
- Justification: All methods sections contain an empty exclusions array and the missing_or_unclear field notes that explicit exclusion criteria are not stated. Without clear criteria, readers cannot assess data quality or potential bias introduced by post-hoc participant removal.

Evidence:

- methods/exp_1 -- field exclusions: [] (no exclusions listed)
- methods/exp_3 -- field exclusions: [] (no exclusions listed)
- methods/exp_4 -- field exclusions: [] (no exclusions listed)

Counterbalancing procedures are insufficiently described for Experiments 3 and 4, despite within-subjects designs that require it

- Severity: **moderate**
- Justification: While Experiment 1 mentions counterbalancing of response keys, Experiments 3 and 4 explicitly note that counterbalancing was not reported. Given their fully within-subjects factorial designs, lack of detail raises concerns about order effects and potential confounds.

Evidence:

- methods/exp_1 -- field counterbalancing: Response key mapping counterbalanced across participants
- methods/exp_3 -- field counterbalancing: not reported
- methods/exp_4 -- field counterbalancing: not reported

No preregistration information is provided for any of the experiments

- Severity: **minor**
- Justification: All methods sections list preregistration_status as 'not reported', indicating that the studies were not pre-registered or that this information was omitted, which limits transparency about analytic decisions.

Evidence:

- methods/exp_1 -- field preregistration_status: not reported
- methods/exp_3 -- field preregistration_status: not reported

- methods/exp_4 -- field preregistration_status: not reported

Reliability estimates for the eye-movement coding (e.g., gaze transition counts) are absent, and coding procedures lack detail

- Severity: **moderate**
- Justification: The studies rely heavily on eye-tracking metrics, yet the methods do not report inter-rater reliability, coding rules, or validation of the gaze transition counts, which undermines confidence in the measurement precision.
- Evidence:
 - methods/exp_1 -- field outcome_measures: gaze transition frequency operationalized but no reliability reporting
 - results_and_discussion/exp_1 -- field missing_or_unclear: No detailed description of the coding procedures for gaze transitions.
 - results_and_discussion/exp_3 -- field missing_or_unclear: No explicit mention of reliability estimates for the measures used.

7. Statistical Robustness

- **Post-hoc pairwise comparisons are reported without clear multiple-comparison corrections, risking inflated Type I error rates.**
- Severity: **moderate**
- Justification: The authors note that "post-hoc Bonferroni-corrected t-tests" were used "where appropriate," yet the reported t-tests lack any correction annotation. Conducting several pairwise tests increases the chance of false positives, especially given the number of outcome measures (RTs, accuracy, gaze transitions). Without explicit correction, the reported significant effects may be overstated.
- Evidence:
 - results_and_discussion/exp_1 -- field post-hoc t-tests: Several t-tests (e.g., compatibility effect for main effects and interactions) are reported with p-values but no indication that Bonferroni or other correction was applied.
 - results_and_discussion/exp_2 -- field post-hoc t-tests: Multiple t-tests comparing compatible vs. incompatible graphs for main effects and interactions are presented (e.g., $t(15)=1.95$, $p=.035$) without specifying correction for the family of tests.
 - results_and_discussion/exp_2 -- field post-hoc t-tests: Additional t-tests for gaze transitions (e.g., $t(15)=4.95$, $p<.001$) are reported without correction despite multiple dependent variables being examined.

